

三、AutoSkill-NL-RL [1 RA], XSY

2025 年 10月16日, Antropic 发布 Agent skill比较明确的定义和规范。但目前构建 skills 仍然有较
大的人
工参与, 以及一个不好的 skill 会导致 agent 性能下降。
openclaw 等智能体在面对复杂任务时, 消耗 token 严重。

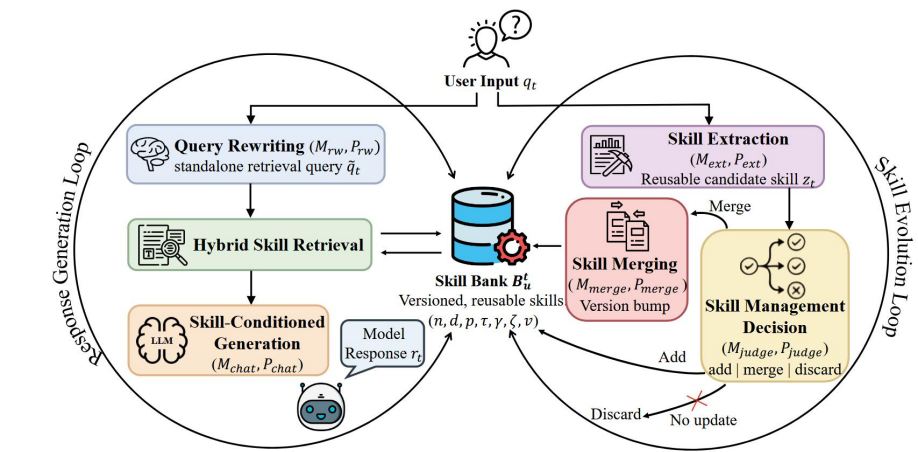


Figure 1: The Framework for our AutoSkill, which is composed of two tightly coupled processes. The right loop, *skill evolution*, transforms interaction experience into explicit skills through extraction and maintenance. The left loop, *skill-enhanced response generation*, uses the current skill bank to support response generation via query rewriting, skill retrieval, and context injection. In this way, the system continually improves through explicit memory growth rather than through model fine-tuning.

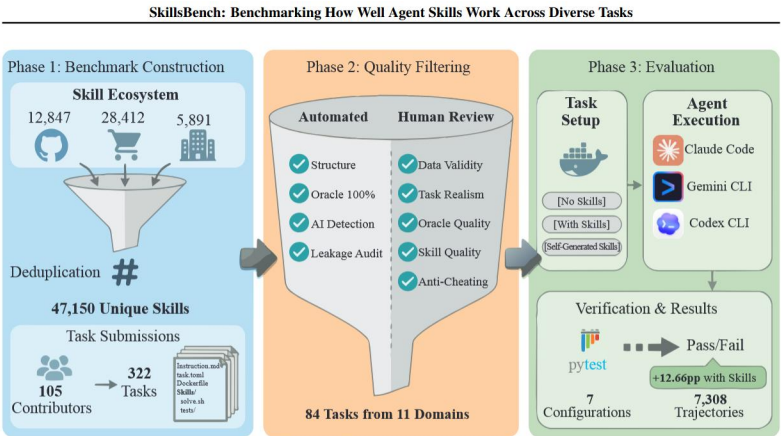
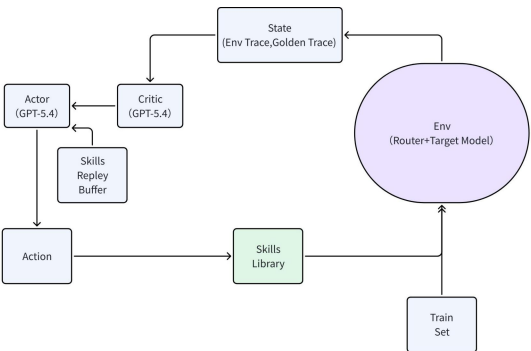


Figure 2. SKILLSBENCH pipeline overview. **Phase 1 (Benchmark Construction):** We aggregate Skills from three sources—open-source repositories (12,847), the Claude Code ecosystem (28,412), and corporate partners (5,891)—yielding 47,150 unique Skills after deduplication. In parallel, 322 contributors submit 105 candidate tasks. **Phase 2 (Quality Filtering):** Each task undergoes automated checks (structural validity, AI detection, leakage audit) and human review (data validity, task realism, oracle quality, Skill quality, anti-cheating), producing 84 tasks spanning 11 domains. **Phase 3 (Evaluation):** Tasks are executed under three conditions (no Skills, with curated Skills, self-generated Skills) across three commercial agent harnesses (Claude Code, Gemini CLI, Codex CLI). Deterministic `pytest` verifiers produce pass/fail outcomes; 7 agent-model configurations yield 7,308 trajectories, with curated Skills providing +12.66pp average improvement.

基于此, 在 MMD 和 Earth agent 的项目中, 发现对于小模型和特定领域固定复杂任务可以有较好的算法
自动构建 skill。



之后的任务：

1. 调试 NL-RL 的框架算法
2. 找寻多个合适的特定领域任务进行评测（Earth Bench、Med）
3. 与 AutoSkill, EvoSkill, Manus 等进行对比